

Programming with C - Assignment1

Total Marks: 100

Due Date: 27/02/2026

Question 1: Student Grade Management System (25 Points)

Write a C program that:

- Reads a student's information:
 - Name (string)
 - Student ID (integer)
 - Scores for 3 subjects (float)
- Calculates the average score
- Uses relational operators to determine the grade:
 - A: average ≥ 90
 - B: average ≥ 80 and average < 90
 - C: average ≥ 70 and average < 80
 - D: average ≥ 60 and average < 70
 - F: average < 60
- Uses logical operators to check if the student passed all subjects (score ≥ 50 in each)
- Displays all information in a formatted manner

Question 2: Circle and Rectangle Calculator (25 Points)

Write a C program that:

- Declares constants for PI (3.14159) using the `const` keyword
- Reads from the user:
 - Radius of a circle (float)
 - Length and width of a rectangle (float)
- Performs calculations using arithmetic operators:
 - Circle: Area = $\pi \times \text{radius}^2$, Circumference = $2 \times \pi \times \text{radius}$
 - Rectangle: Area = length $\times$ width, Perimeter = $2 \times (\text{length} + \text{width})$
- Uses the ternary operator to determine which shape has the larger area
- Displays all results with 2 decimal places

Question 3: Number Operations Program (25 Points)

Write a C program that:

- Reads two integers from the user (a and b)
- Performs and displays:
 - All arithmetic operations: +, -, *, /, % (check for division by zero)
 - Comparison results using relational operators:
 - Is a equal to b?
 - Is a greater than b?
 - Is a less than or equal to b?
 - Logical operations:
 - Are both numbers positive? (use &&)
 - Is at least one number even? (use || and %)
 - Pre-increment and post-increment demonstration on variable a
 - Bitwise AND, OR, and XOR operations on both numbers

Question 4: Temperature and Time Converter (25 Points)

Write a C program that:

- Reads temperature in Celsius (float) and time in seconds (integer) from the user
- Converts temperature to:
 - Fahrenheit: $F = (C \times 9/5) + 32$
 - Kelvin: $K = C + 273.15$
- Converts time in seconds to hours, minutes, and remaining seconds using arithmetic operators (/ , %)
- Uses the ternary operator to classify temperature:
 - "Hot" if Celsius > 30
 - "Moderate" if Celsius between 15 and 30
 - "Cold" if Celsius < 15
- Uses logical operators to check if:
 - Temperature is within a comfortable range (18°C - 25°C)
 - Time is more than 1 hour (3600 seconds)
- Displays all conversions and checks